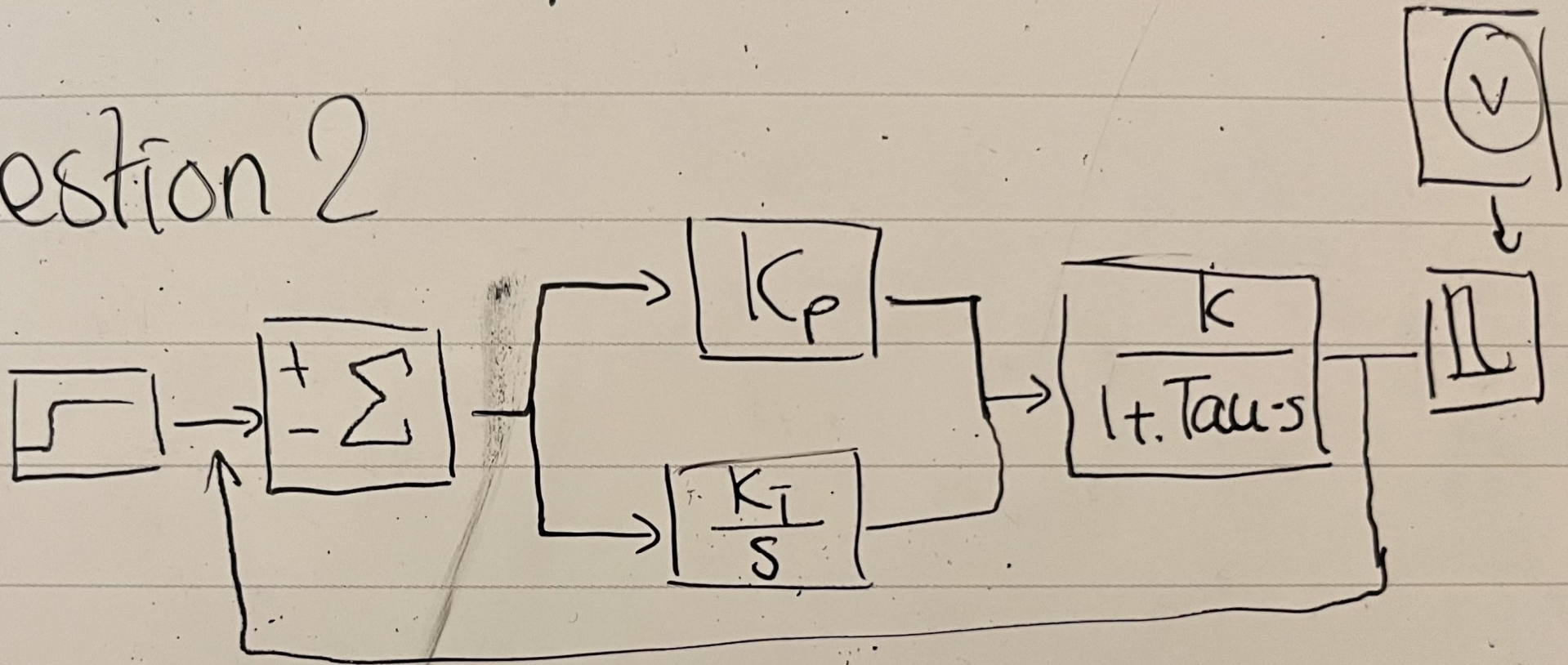


Question 2

a.



b. Question 2

$$H_1 = K_p \cdot \frac{k_i}{s} \cdot \frac{k}{1 + \tau_{au} s}$$

$$= \frac{K_p \cdot k_i \cdot k}{s(1 + \tau_{au} s)}$$

$$H_1 = 1;$$

$$H_1 = \left(\frac{K_p \cdot k_i \cdot k}{s(1 + \tau_{au} s)} \right) \cdot \left(\frac{K_p \cdot k_i \cdot k}{s(1 + \tau_{au} s)} \right) \cdot \left(\frac{K_p \cdot k_i \cdot k}{s(1 + \tau_{au} s)} \right)$$

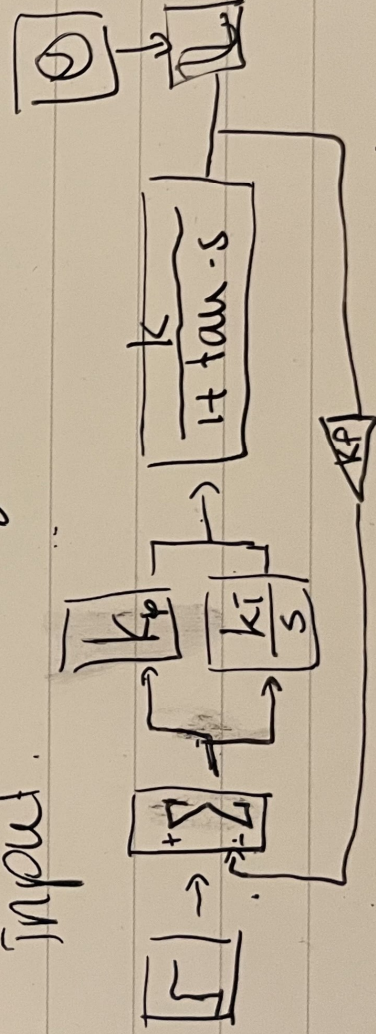
$$= \left(\frac{K_p \cdot k_i \cdot k}{s(1 + \tau_{au} s)} \right) \cdot \left(\frac{K_p \cdot k_i \cdot k}{s(1 + \tau_{au} s)} \right) \cdot \left(\frac{K_p \cdot k_i \cdot k}{s(1 + \tau_{au} s)} \right)$$

$$= \left(\frac{K_p \cdot k_i \cdot k}{s(1 + \tau_{au} s)} \right) \cdot \left(\frac{K_p \cdot k_i \cdot k}{s(1 + \tau_{au} s)} \right) \cdot \left(\frac{K_p \cdot k_i \cdot k}{s(1 + \tau_{au} s)} \right)$$

$$= \frac{s(1 + \tau_{au} s) \cdot (s(1 + \tau_{au} s) + K_p \cdot k_i \cdot k)}{s(1 + \tau_{au} s) + K_p \cdot k_i \cdot k}$$

$$= s(1 + \tau_{au} s)$$

c. there should be a gain added to manage the errors, won't get too big when there is a large difference in input.



Question 1

a. the output is 75, the setpoint is 150. meaning

$$kf \rightarrow 150 = 75 \cdot kf$$

b. during rest the heater is room temperature ($\pm 20^\circ\text{C}$) $\rightarrow$ meaning the feedback is

$$20 \cdot 2 = 40$$

the error would be the setpoint (150) - the feedback

$$\rightarrow \text{error} = 110$$

taking the worst case

scenario where the input volt for the heater is

230; the k_p would be

$$230:110 \approx 2.1$$

$$c \quad H_1 = 2 \cdot \frac{8}{245+0.5}$$

$$\begin{aligned}
 H_1 &= \left(\frac{8}{245+0.5} \right) / \left(\left(\frac{8}{245+0.5} \right) / \left(1 + \frac{8}{245+0.5} \cdot 2 \right) \right) \\
 &= \frac{8}{245+0.5} / \left(\left(\frac{8}{245+0.5} \right) / \left(1 + \frac{32}{485+1} \right) \right) \\
 &= \left(\frac{8}{245+0.5} \right) / \left(\left(\frac{8}{245+0.5} \right) / \left(\frac{485+33}{485+1} \right) \right) \\
 &= \left(\frac{8}{245+0.5} \right) / \left(\frac{8}{245+0.5} \cdot \frac{2(245+0.5)}{485+33} \right) \\
 &= \frac{8}{245+0.5} / \left(8 \cdot \frac{2}{485+33} \right)
 \end{aligned}$$

Question 1

$$\begin{aligned}
 C \quad H_1 &= 2 \cdot \frac{24s+0,5}{8} = \frac{24s+0,5}{8} \\
 H_f &= \left(\frac{24s+0,5}{8} \right) / \left(\frac{24s+0,5}{8} \right) / \left(1 + \frac{24s+0,5}{32} \cdot 2 \right) \\
 &= \left(\frac{24s+0,5}{8} \right) / \left(\frac{24s+0,5}{8} \right) / \left(1 + \frac{48s+1}{32} \right) \\
 &= \left(\frac{24s+0,5}{8} \right) / \left(\frac{24s+0,5}{8} \right) / \left(\frac{48s+33}{2(24s+0,5)} \right) \\
 &= \left(\frac{24s+0,5}{8} \right) / \left(\frac{24s+0,5}{8} \cdot \frac{48s+33}{48s+33} \right) \\
 &= \left(\frac{24s+0,5}{8} \right) / \left(\frac{8 \cdot (48s+33)}{16} \right) \\
 &= \left(\frac{24s+0,5}{8} \right) / \left(\frac{48s+33}{2} \right) \\
 &= \frac{24s+0,5}{8} \cdot \frac{2}{48s+33} \\
 &= \frac{24s+0,5}{48s+33} \\
 &= \frac{1}{48s+33}
 \end{aligned}$$

